Computational Analysis of 1,382 Jazz Standards Using K-Means Clustering

1. Project Overview and Problem Statement

The primary objective of this project is to discover the stylistic patterns of jazz standards through unsupervised learning. Traditional musicology often focuses on a few dozen canonical tracks, leaving the broader stylistic shifts of the 20th century under-quantified. By processing the harmonic and rhythmic varieties in 1,382 jazz standards, we aim to uncover the underlying structures that define the different eras and styles of jazz, from traditional to modal.

2. Data Wrangling

In our analysis, we transformed a raw JSON file of 1,382 jazz standards into a high-dimensional feature set ready for unsupervised clustering.

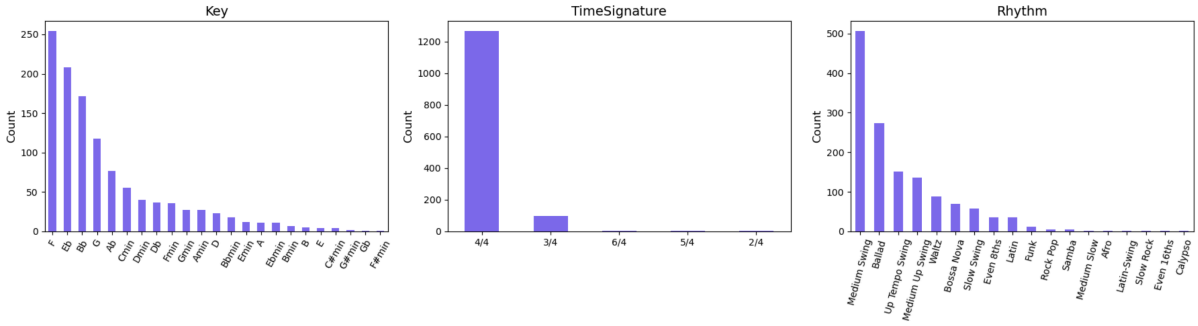
- **Data Source:** The primary dataset originated from the JazzStandards repository by Mike Oliphant - a raw JSON file (JazzStandards_raw.json) containing 1,382 standards, characterized by nested sections and chords.
- **Data flattening:** To analyze harmonic progressions, the JSON was flattened into a one-row-per-chord format, resulting in a dataset of 56,025 individual chord entries.
- **Composition year** was added to each jazz standard through a combination of manual and GPT-assisted internet search.

3. Exploratory Data Analysis

Initial exploratory analysis provided an initial glimpse at the harmonic and rhythmic diversity in the dataset.

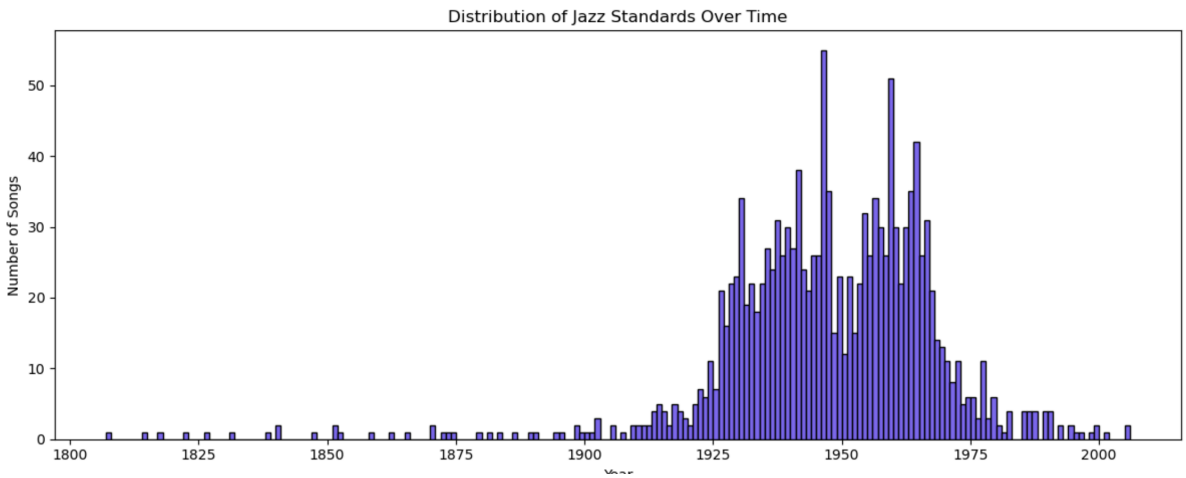
a) Stylistic Trends

The repertoire is characterized by a strong preference for horn keys like F, Eb, and Bb, likely due to their ease for wind instruments. 4/4 time and Medium Swing or Ballad are the dominant rhythms. Newer styles such as Bossa Nova, Funk, and Rock/Pop emerge after the mid-1950s.



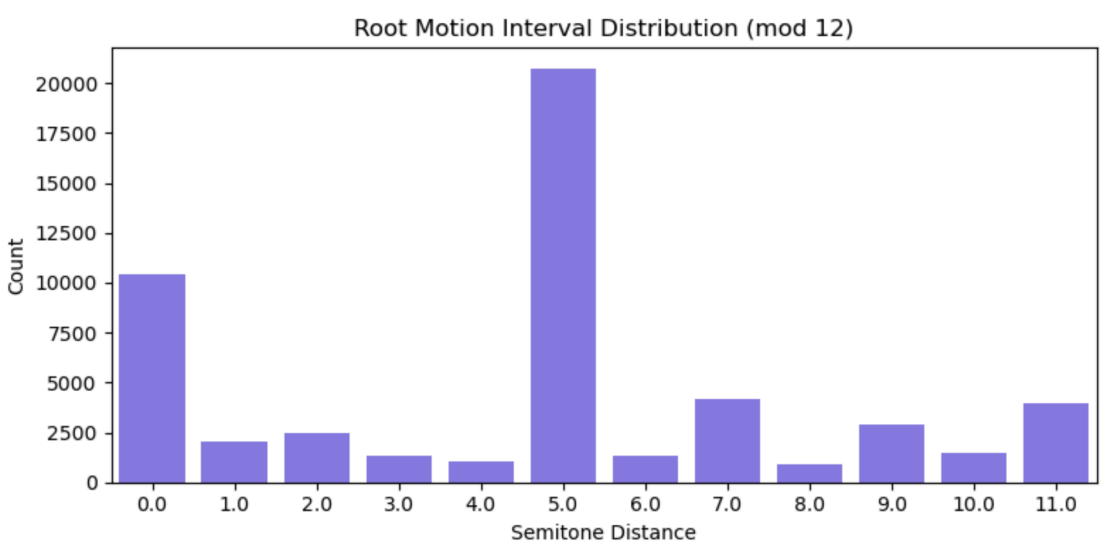
b) Composition Year

Composition years span the 20th century but show a significant peak during the 1950s and 1960s, marking the most prolific era for jazz standards.



c) Root Motion

The dataset confirms the dominance of the ii–V–I progression in jazz harmony, finding that 39.29% of all root motion moves up a perfect fourth (or down a fifth). Dominant 7th chords are the most frequent chord type (21,589 instances). While extensions like 9ths and 6ths are common for color, the majority of lead-sheet chords are noted in their base form.



4. Feature Engineering and Pre-Processing

To prepare for unsupervised clustering, we engineered features to capture the harmonic footprint of each composition:

- **Harmonic Density:** Measured as chords per measure; higher density indicates a faster harmonic rhythm typical of Bebop.
- **Harmonic Vocabulary:** Relative distribution of Triads, Maj7, m7, and Dominant 7th chords.
- **Root Motion:** Defined by the percentage of "descending fifths" (root motion moving up 5 half-steps/down a fifth). This movement accounts for 39.29% of all root motions, highlighting the famous 2-5-1 progression.
- **Extensions:** The percentage of "9plus" extensions (6ths, 9ths, 11ths, and 13ths).
- **Tonal Entropy:** Using scipy.stats.entropy to calculate the unpredictability of pitch classes. This distinguishes the repetitive, diatonic nature of early standards from the more chromatic nature of modern jazz.
- **Relevant Metadata**: Time signature, year, key, and rhythm style. Categorical features were converted into binary dummy variables using pd.get_dummies.

5. Modeling: Unsupervised Clustering

We evaluated three clustering algorithms to identify natural stylistic groupings. The elbow method and silhouette score were utilized to determine that k=4 provided the optimal balance of granularity and interpretability.

a) Model Selection and Evaluation

Models were assessed using Silhouette Scores to measure cluster cohesion and separation.

Model	Train Silhouette Score	Test Silhouette Score
K-Means (k=4)	**0.1363**	**0.1366**
GMM (k=2)	0.1699	0.1099
Agglomerative	0.0697	0.0981

Decision: KMeans was selected as the final model due to its stability across unseen data. While GMM showed a higher training score, its sharp drop on the test set suggested overfitting. PCA Projection confirmed that while stylistic boundaries in music are naturally overlapping, KMeans provided the most consistent partitions for stylistic analysis.

6. Core Findings: The Four Clusters of Jazz

The model identified four distinct stylistic profiles that align with the genre's historical evolution.

Cluster 1: Harmonically Dense Traditional Standards

- **Mean Year:** 1943
- **Harmonic Density:** 4.89 chords per measure (highest in dataset).
- **Root Motion:** 59.3% descending fifths (traditional ii-V-I motion).
- **Keys:** Mostly Eb (21.4%) and F (20.8%).
- **Rhythm:** 4/4 time; Medium Swing (42%) and Ballads (32%).

Cluster 2: Dominant-Heavy / Harmonically Simple Standards

- **Mean Year:** 1942
- **Chord Type:** 55.1% Dominant 7th chords.
- **Extensions:** Low use of complex harmony (only 7.8% 9+ extensions).
- **Keys:** Concentrated in F (19.6%) and Bb (18.6%).
- **Rhythm:** Centered on Medium Swing (42.8%).

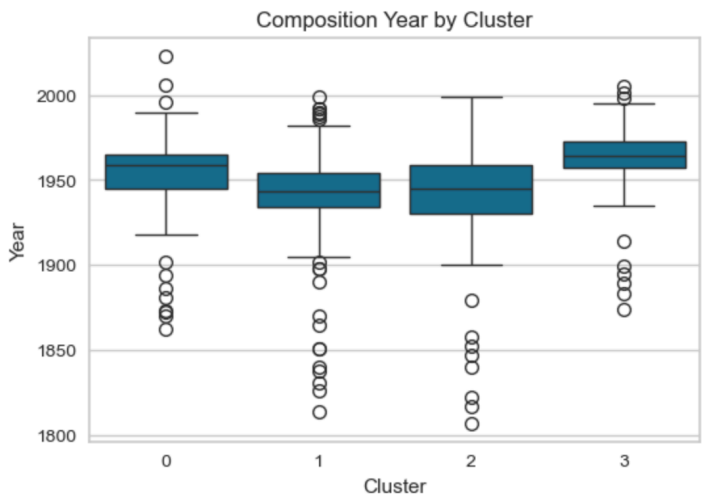
Cluster 0: Rhythmically Diverse / Mid-Century Standards

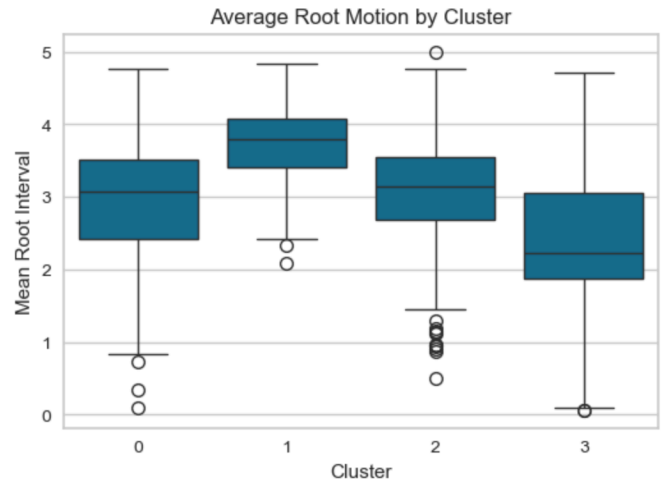
- **Mean Year:** 1954
- **Style:** High prevalence of Bossa Nova (14.1%) and Waltz (15.1%).
- **Harmonic Density:** 2.24 chords per measure.
- **Keys:** Reliance on F (15.8%) and Eb (11.3%).

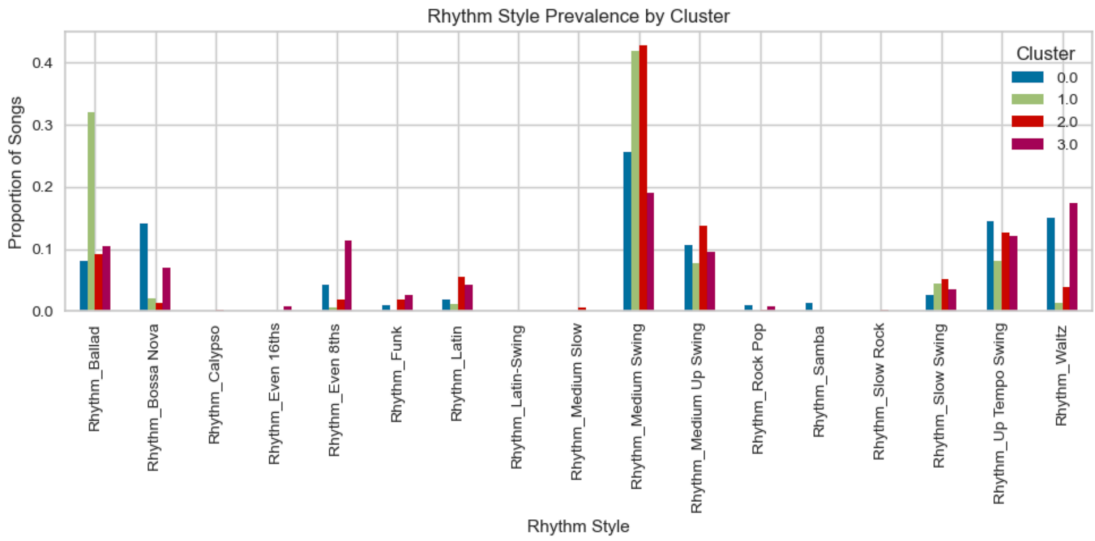
Cluster 3: Post-Bop / Modal / Later Standards

- **Mean Year:** 1962
- **Extensions:** Highest use of 9+ extensions (47.7%).
- **Root Motion:** Only 27.7% descending fifths (move away from traditional harmony).
- **Time/Key:** 18.3% Waltz time (3/4). Diverse keys like Cmin, Gmin, and Db.

These four clusters reveal distinct harmonic and stylistic groupings within jazz standards, corresponding to different historical periods and harmonic trends.







7. Further Research

While this project focused exclusively on jazz standards, the same computational framework could be expanded to include other popular music traditions such as pop, rock, and folk. Incorporating chord charts from these genres would allow for cross-genre clustering to see whether the harmonic patterns that define jazz also appear in other forms of popular music.

Future work could also explore alternative musical features. For example, incorporating tempo, song form, and roman numeral analysis could reveal further stylistic distinctions.